

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

WINTER SEMESTER, 2023-2024
FULL MARKS: 150

CSE 4501: Operating Systems

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 5 (five)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

Table 1: System overview for Question 1.

Process	MAX			ALLOC			REQ			AVAIL		
	A	B	C	A	B	C	A	B	C	A	B	C
P_1	6	6	5	5	2	3	0	0	2			
P_2	4	4	3	1	1	3	2	0	0			
P_3	3	6	5	3	3	2	1	1	1			
P_4	3	4	6	2	3	4	0	0	1	4	4	2
P_5	2	3	4	0	1	1	2	1	1			
P_6	4	5	2	0	1	0	2	1	1			

- Consider the system overview as shown in Table 1, which consists of 6 processes (P_1 through P_6) and 3 types of resources A, B, and C with 15 instances each. Each process in the system has the parameters — *maximum number of resources required* (MAX), *resources currently allocated* (ALLOC), and *number resources requested* (REQ). The system constantly monitors the currently available (AVAIL) resources as well as those already allocated (ALLOC) to keep the system safe from a deadlock situation. Based on this information, answer the following —
 - For each of the schemes, *deadlock-avoidance* and *deadlock-detection*, devise respective algorithms and determine whether the system currently is in a safe state. Provide step-by-step illustrations for each algorithm. 10 + 10
(CO3)
(PO2)
 - If the system only employs the *deadlock-detection* scheme, how frequently should the corresponding algorithm be invoked? Elaborate on the policies that should be in place to recover from the deadlock as detected by the *deadlock-detection* algorithm. 5 + 10
(CO2)
(PO1)
- Consider a computer system that runs 10000 jobs per month with no *deadlock-avoidance* scheme. About 100 deadlocks occur monthly, and the operator must terminate and rerun about 20 jobs per deadlock. Each job is worth about \$6.00 (in CPU time), and the jobs terminated tend to be about half-done when aborted. A system programmer has estimated that a *deadlock-avoidance* algorithm could be installed in the system with an increase in the average execution time per job of about 15%. Since the machine currently remains idle 30% of the time, all 10000 jobs per month could still be run, although turnaround time would increase by about 20% on average. Based on this information, answer the following —
 - Calculate the net change in monthly revenue before and after installing the *deadlock-avoidance* algorithm. 20
(CO3)
(PO2)
 - Discuss whether installing the *deadlock-avoidance* algorithm is financially justified. 5

Table 2: System overview for Question 3.

Process	Arrival Time	Priority	Burst Time
P_1	0	2	25
P_2	5	1	5
P_3	10	3	10
P_4	14	4	20
P_5	5	5	15

3. Consider the system overview, as shown in Table 2, which lists the *arrival times*, *priorities*, and *burst times* of 5 processes (P_1 through P_5). The system also has *interrupt* and *dispatch* latencies of 1 ms each. Based on this information, answer the following —
- Before P_1 arrived, suppose the system was executing the process P_0 , a low-priority process with a burst time of 100 time units (not listed in the table). Considering the process execution timeline resets with the arrival of P_1 (arrival time is at zero), which of the two algorithms, *Shortest Remaining Time First (SRTF)* and *Preemptive Priority Scheduling (PPS)* will result in a *reduced average waiting time* for the processes in Table 2? Your answer must include the Gantt charts for each algorithm. 10 + 10
(CO2)
(PO2)
 - Compute the average turnaround time of the processes if the CPU after remaining idle for the first 15 time units employs the *Round Robin* scheduling algorithm with time quantum $Q = 5$. 10
(CO2)
(PO2)
 - Use the *Round-Robin* scheduling algorithm to analyze the impact of a small quantum value, $Q = 2$, compared to a high value, $Q = 10$, on the average waiting time of the processes in Table 2. 10
(CO2)
(PO2)
4. a) The traditional UNIX scheduler enforces an inverse relationship between priority numbers and priorities. The scheduler recalculates priority of any process once per second using the following formula —
- $$\text{priority} = (\text{recent_CPU_usage} / 2) + \text{base}$$
- Here, $\text{base} = 60$ and recent_CPU_usage indicates how often a process has used the CPU since priorities were last recalculated. Assume that recent_CPU_usage for three processes, P_1 , P_2 , and P_3 are 40, 18, and 10, respectively. What will be the new priorities for these three processes when priorities are recalculated? Based on this information, does the traditional UNIX scheduler raise or lower the relative priority of a CPU-bound process? 6
(CO2)
(PO2)
- b) The fourth and final condition for deadlocks is the circular-wait condition. Considering the *deadlock-prevention* scheme, devise two protocols to prevent this condition from occurring, with proper illustrations, and prove their validity using proof by contradiction. 10 + 5
(CO3)
(PO2)
5. a) What is *busy waiting*? Does an operating system provide any alternative to it? If yes, describe its mechanism. 1 + 3
- b) Elaborate on how the issues related to — the data representation techniques between client and server, the semantics of a procedure call, and establishing communication between a server and a client, are handled while implementing a Remote Procedure Call (RPC) mechanism. 4 × 3
- c) Define *sockets*. Write a JAVA program that allows a client to request the current date and time from a date server, listening to port 6013, while using connection-oriented TCP sockets. It is to be noted that the client and the server are on the same machine and when the latter receives a request from the former, it should return the corresponding date and time. Provide a timeline diagram of a sample execution of your program. 1 +
10 + 2
(CO2)
(PO1)

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CSE 4503: Microprocessor and Assembly Language

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Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

-
1. a) What was the alternative of Interrupt? Explain and classify different types of Interrupt. 3 + 6
(CO1)
(PO1)
 - b) What is Duty Cycle? Find out the relation between Clock State, Bus/Machine Cycle and Instruction Cycle. 3 + 6
(CO1)
(PO1)
 - c) Compare and contrast between the control signal operations of Maximum and Minimum modes of 8086 microprocessor. 7
(CO4)
(PO2)
 2. a) Draw the details of WRITE Bus timing diagram showing all the necessary signals of 8086. You should consider that the write operation is to be made to an output device with port address of 0FFFFh. Consider that there is 4 WAIT states, interrupt flag is SET and DS register is in use for data accessing. 9
(CO4)
(PO2)
 - b) What is MSW register? Explain the functionalities of IOPL, NT, and VM flags. 3 + 6
(CO1)
(PO1)
 - c) Suppose, while debugging an assembly language program the values of the registers are: Flag=ABCDh, IP=0102h, CS=8000h, SP=FFFAh. Now, if INT 21h is requested, derive the memory addresses from where the new IP and CS can be retrieved. Also, show the new SP value and steps involved in handling the interrupt by the 8086 microprocessor. 7
(CO4)
(PO2)
 3. a) How can multi-tasking be ensured with protected mode of a microprocessor? Differentiate between Real-mode, Protected Mode, and VM8086 Mode. 3 + 6
(CO2)
(PO2)
 - b) Draw the structures of memory banks of 80286, 80386, and Pentium. Mention how the address pins are used to identify the memory banks. 3 + 6
(CO3)
(PO1)
 - c) "A Pentium processor is 5 times faster than the 80486 numeric processor" – Justify the statement with an appropriate example. 7
(CO3)
(PO1)
 4. a) What is memory segment overlapping? Write short notes on logical address, linear address, and physical address. 3 + 6
(CO3)
(PO1)
 - b) What is Paging? Describe the process of calculation of linear address while paging is enabled and disabled. 3 + 6
(CO2)
(PO1)
 - c) Suppose, you are given a memory address of 4GB (with addresses 00000000h to FFFFFFFFh). Using a 16-bit segment register, clearly explain the process to define a global descriptor table along with local descriptor table. 7
(CO4)
(PO2)

5. a) Drawing the architectures differentiate between CPU and GPU? 9
(CO3)
(PO1)
- b) "GPU is designed by optimizing few traditional CPU components" - What are those components and how are those optimized in GPU? 9
(CO3)
(PO1)
- c) Discuss the "Single/Multiple Instruction Stream and Single/Multiple Data Stream" considerations for designing single/multi-core processors. 7
(CO3)
(PO3)
6. a) Write short notes on the followings using appropriate example/figure: 5 × 5
(CO1)
(PO1)
- i. Thread
 - ii. Hyper-thread
 - iii. Turbo-boost
 - iv. Scalar Processor
 - v. CUDA Framework

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CSE 4511: Computer Networks

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Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) An organization is granted a class B (172.16.0.0) and a class A (10.0.0.0) network address. The administrator wants to apply the subnet mask 255.255.255.224 to create subnets from both of the given networks. Answer the following questions for both of the networks: 10
(CO2)
(PO1)
 - i. How many subnets can there be?
 - ii. Find the number of usable addresses in each subnet.
 - iii. Find the first usable address, the last usable address, and the broadcast address on the first subnet.
 - iv. Find the first usable address, the last usable address, and the broadcast address on the last subnet.
- b) An IP datagram of size 3000 byte is sent out from the Source device to Destination device depicted in Figure 1. Due to differences in MTU in the physical network of each LAN this datagram could be fragmented. If fragmentation takes place, the first fragment follows the path Source - LAN 1 - LAN 2 - LAN 5 - Destination. The rest of the fragments follow the path Source - LAN 1 - LAN 3 - LAN 4 - LAN 5 - Destination. What are the fragmented datagrams received at the destination? Include all the header information in your answer which is required for the reassembly in the destination for each fragmented datagram. 15
(CO2)
(PO2)

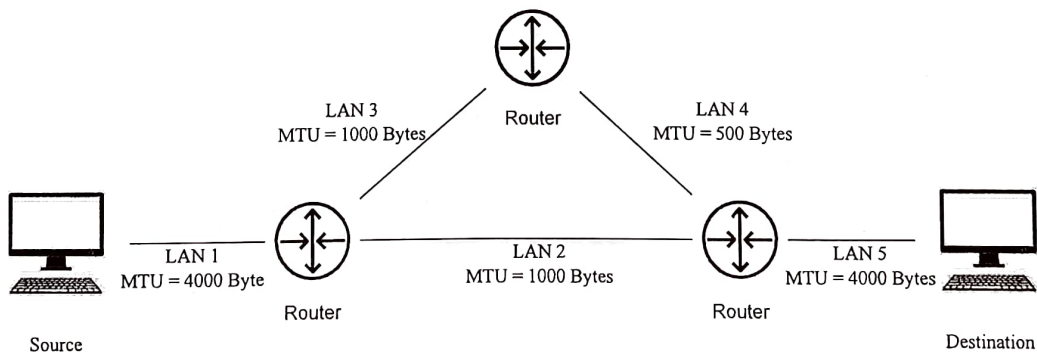


Figure 1: Network topology for Question 1.b

2. a) A MAC address can represent a unicast, multicast, or broadcast address. What types of addresses are the following? 5
(CO1)
(PO1)
 - i. 4A:30:10:21:10:1A
 - ii. 47:20:1B:2E:08:EE
 - iii. FF:FF:FF:FF:FF:FF
- b) Draw the flow diagram of CSMA/CD. Why is there a fixed value of the number of attempts ($k < 15$) present? 7 + 3
(CO2)
(PO1)

- c) The following frames are passed sequentially in the network topology depicted in Figure 2.
- A sends frame to D
 - D sends frame to B
 - B sends frame to A
 - C sends frame to D

10
(CO2)
(PO1)

What will be the status of the forwarding table of the bridge after each event mentioned above? Mention which port the frame will take at each event.

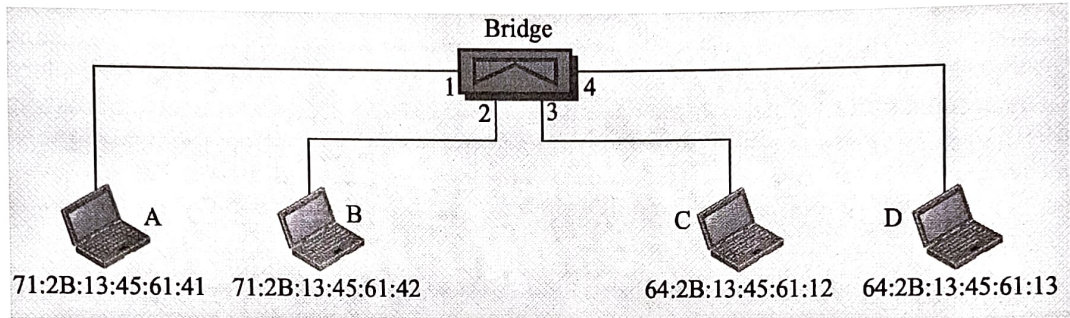


Figure 2: Network topology for Question 2.c

3. a) A TCP client opens a connection using an initial sequence number (ISN) of N (where N:= Last 4 digits of your student ID). The TCP server opens the connection with an ISN of M (where $M:=N+1000$). Draw the three TCP segments during the three-way handshaking connection establishment. (Use timeline in y-axis for each side to show the states and the relative duration of the client and the server.)
- b) The dump of a UDP header in hexadecimal format is

10
(CO2)
(PO1)

$(0045DF00 \ 0058FE20)_{16}$

5
(CO2)
(PO1)

Answer the following:

- What is the source port number?
 - What is the destination port number?
 - What is the total length of the user datagram?
 - What is the length of the data?
 - Is the packet directed from a client to a server or vice versa?
 - What is the client process?
- c) The dump of a TCP header in hexadecimal format is

$(05320017 \ 00000001 \ 00000000 \ 500207FF \ 00000000)_{16}$

5
(CO2)
(PO1)

Answer the following:

- What is the source port number?
 - What is the destination port number?
 - What the sequence number?
 - What is the acknowledgment number?
 - What is the length of the header?
 - What is the type of the segment?
 - What is the window size?
- d) What is a Transmission Control Block (TCB)? What are the different fields of TCB?

5
(CO2)
(PO1)

4. a) TCP uses sliding window technique for flow control. The sender has sent bytes upto 202 and the receiver has sent an acknowledgment number of 200 with an rwnd of 12 bytes (assume the cwnd is 20).

10
(CO2)
(PO1)

- Depict the current window. After some time, the sender has received an acknowledgment value of 204 and an rwnd of 6.
- Draw the new window.
- Is there any problem with the current window? If yes, explain the problem with the possible solution.

- b) The graph in Figure 3 depicts the cwnd vs. RTTs of a TCP connection. Based on the graph and congestion policy of TCP, describe the possible events with detail explanation.

15
(CO2)
(PO2)

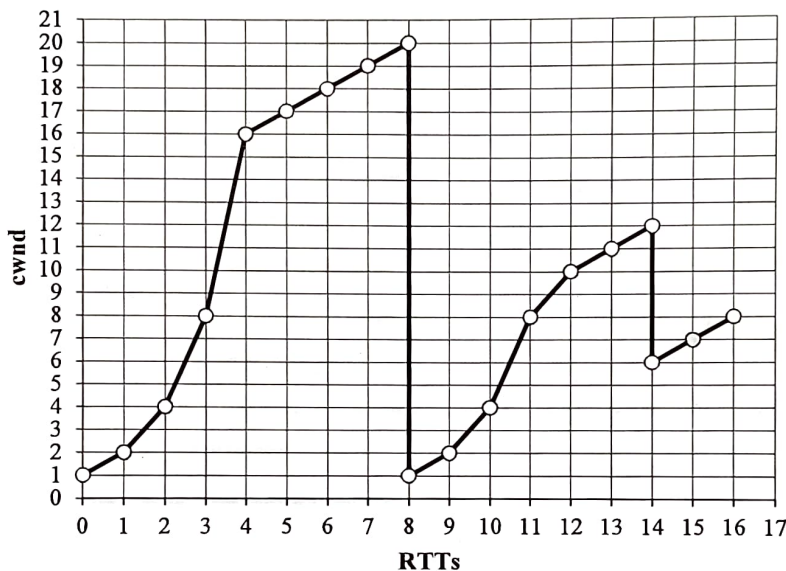


Figure 3: A graph of cwnd vs RTTs for Question 4.b

5. a) The dump of a DATA chunk in hexadecimal format is

(00000015 00000005 0003000A 00000000 48656C6C 6F000000)₁₆

10
(CO2)
(PO1)

Answer the following questions (use decimal value in the answer).

- Is this an ordered or unordered chunk?
- Is this the first, the last, the middle, or the only fragment?
- How many bytes of padding are carried by the chunk?
- What is the TSN?
- What is the SI?
- What is the SSN?
- What is the message?

- b) An SCTP client opens an association using an initial tag of 2200, an initial TSN of 11111, and a window size of 30000. The server responds with an initial tag of 1100, an initial TSN of 250, and a window size of 15000. Show the time-line diagram of the association establishment. Your answer should include the all contents of SCTP packet which are available. What is the significance of cookie in association establishment?

6 + 2
(CO2)
(PO1)

- c) Explain the scenario of simultaneous open operation of SCTP with a labelled timing diagram.

7
(CO2)
(PO1)

6. a) Explain all the security goals of a digital communication? Explain the common threats to security goals. 10
(CO1)
(PO1)

b) The Vigenère cipher is a classical polyalphabetic substitution cipher. It uses a word as keystream. Such as "ADD" key implies that $K_1 = 0$, $K_2 = 3$, $K_3 = 3$. What will be the plain text for the following ciphertext with the key "faith" considering no position dependence is present? 15
(CO1)
(PO1)

sedxy yhqgr yhim hqlia px uvtdfrm hm bhim ame ekvsglhlws lh

Appendix

0	3	4	7	8	15	16	31
VER 4 bits	HLEN 4 bits	Service type 8 bits			Total length 16 bits		
Identification 16 bits					Flags 3 bits	Fragmentation offset 13 bits	
Time to live 8 bits		Protocol 8 bits			Header checksum 16 bits		
Source IP address							
Destination IP address							
Options + padding (0 to 40 bytes)							

(a) Header format of IP datagram

1	16	31
Source port address 16 bits		Destination port address 16 bits
Sequence number 32 bits		
Acknowledgment number 32 bits		
HLEN 4 bits	Reserved 6 bits	Window size 16 bits
Checksum 16 bits		Urgent pointer 16 bits
Options and padding		

(b) Header format of TCP Segment

0	16	31
Source port number		Destination port number
Total length		Checksum

b. Header format

(c) Header format of UDP Datagram

0	7	8	15	16	31
Type		Flag		Length	
Chunk Information (multiple of 4 bytes)					

(d) Common layout of a chunk of SCTP

0	7	8	15	16	31
Type: 1		Flag: 0		Length	
Initiation tag					
Advertised receiver window credit					
Outbound streams			Maximum inbound streams		
Initial TSN					
Variable-length parameters (optional)					

(e) Format of INIT Chunk of SCTP

0	7	8	15	16	31
Type: 2		Flag: 0		Length	
Initiation tag					
Advertised receiver window credit					
Outbound streams			Maximum inbound streams		
Initial TSN					
Parameter type: 7			Parameter length		
State cookie					
Variable-length parameters (optional)					

(f) Format of INIT ACK Chunk of SCTP

0	7		8	13		14	15	16	31
Type: 0			Reserved		U	B	E	Length	
Transmission sequence number									
Stream identifier					Stream sequence number				
Protocol identifier									
User data									

(g) Format of Data Chunk of SCTP

Port	Protocol	Description
7	Echo	Echoes a received datagram back to the sender
9	Discard	Discards any datagram that is received
11	Users	Active users
13	Daytime	Returns the date and the time
17	Quote	Returns a quote of the day
19	Chargen	Returns a string of characters
53	Domain	Domain Name Service (DNS)
67	Bootsps	Server port to download bootstrap information
68	Bootpc	Client port to download bootstrap information
69	TFTP	Trivial File Transfer Protocol
111	RPC	Remote Procedure Call
123	NTP	Network Time Protocol
161	SNMP	Simple Network Management Protocol
162	SNMP	Simple Network Management Protocol (trap)

(h) Well known port of UDP Services

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SEMESTER FINAL EXAMINATION

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DURATION: 3 HOURS

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CSE 4513: Software Engineering and Object-Oriented Design**Programmable calculators are not allowed. Do not write anything on the question paper.**

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Suggest three reasons why the productivity rate of programmers working as a pair might be more than half that of two programmers working individually. 6
- b) What are the five fundamental feasibility evaluations crucial for determining a software project's viability, practicality, and strategic coherence before initiation? 5
- c) Why are story points used for estimation in Agile rather than time-based estimates? Explain the difference between a requirement and its specification with the help of an example. 4 + 10
2. Consider a banking application that needs to integrate with three distinct third-party services, each requiring different communication protocols. The first service, *GlobalPay*, uses a *SOAP-based* protocol for communication. The second service, *QuickPay*, operates with a *RESTful API* that communicates in *JSON* format. Lastly, the *Fraud Detection Service* expects data in *XML* format, while the banking application processes and communicates exclusively using *JSON*. Given this scenario, answer the following:
 - a) Identify the most suitable Design Pattern for resolving the integration challenges between the banking application and three third-party services. Construct the corresponding UML class diagram and discuss the role and interaction of each component. 10 (CO2) (PO2)
 - b) Explain how the chosen design pattern adheres to the SOLID principles. Discuss how each of the principles is applied in the context of this solution, focusing on maintainability, scalability, and flexibility. 5 + 10 (CO2) (PO2)
3. a) A software development company is designing an **Enterprise Resource Planning (ERP)** system consisting of multiple interconnected components that rely on each other to ensure seamless functionality. The system includes a **User Management (UM)** module responsible for authentication and access control, which is utilized by the **Order Processing (OP)** module to verify customer identities before transactions and by the **Notification System (NS)** to send security alerts. The **Inventory Management (IM)** module, which keeps track of stock levels and supplier information, is accessed by both **OP** for order fulfillment and the **Reporting Module (RM)** for generating inventory reports. The **OP** module, which handles customer orders, payments, and invoicing, depends on **IM** for stock verification and also interacts with **RM** to generate sales reports. Additionally, **RM** relies on both **UM** and **IM** to compile detailed analytics, while **NS** is triggered by **OP** to send payment confirmations and shipping updates. Given this scenario, answer the following:
 - i. Calculate the Instability (I) for each component and find the most stable and most unstable components in the system.
 - ii. How does the stability of components impact system maintainability and design?
- b) In a large software system, circular dependencies between modules like Payment Processing, Order Management, and Notification Services make maintenance and updates difficult. How can the **Acyclic Dependencies Principle (ADP)** help fix this? Explain possible solutions and their trade-offs. 10 (CO2) (PO2)

4. a) Consider the following code snippet:

5 + 10

```
1  #include <stdio.h>
2  #define MAX_BALANCE 10000
3  typedef struct {
4      int account_number;
5      char name[50];
6      double balance;
7  } BkAcc;
8  int withdraw_money(BkAcc* account, double amount) {
9      if (account->balance - amount < 0) {
10         return -1;
11     }    account->balance -= amount;
12     return 0;
13 }
14 int transfer_money(BkAcc* from, BkAcc* to, double amount) {
15     if (from->balance < amount) {
16         return -2;}
17     from->balance -= amount;
18     to->balance += amount;
19     return 0;
20 }
21 void print_balance(BkAcc account) {
22     printf("Account %d - Name: %s - Balance: %.2f\n", account.
        account_number, account.name, account.balance);
23 }
24 void process_transaction(BkAcc* account, double withdrawal, double
    transfer, BkAcc* target) {
25     if (withdraw_money(account, withdrawal) == -1) {
26         printf("Error: Insufficient funds for withdrawal.\n");
27     }
28     if (transfer_money(account, target, transfer) == -2) {
29         printf("Error: Insufficient funds for transfer.\n"); }
30 }
31 int main() {
32     BkAcc account1 = {1001, "John Doe", 5000};
33     BkAcc account2 = {1002, "Jane Smith", 1500};
34     double withdrawal_amount = 6000;
35     double transfer_amount = 2000;
36     process_transaction(&account1, withdrawal_amount, transfer_amount
        , &account2);
37     print_balance(account1);
38     print_balance(account2);
39     return 0;
40 }
```

Code Snippet 1: Code Snippet for Question 4.a

- i. Identify potential violations of clean coding principles present in the code snippet.
 - ii. Provide suggestions for each of the identified violations to make the code easier to understand and maintain.
- b) With an example, illustrate the different combined fragments that can be present in a sequence diagram.

10

5. a) A program takes an integer as input, and if the input is a prime number, the program repeatedly sums its digits until the result is a single digit. For this program, identify all the independent paths and design test cases based on those paths. 10
(CO3)
(PO2)
- b) You are testing a mobile banking application that allows users to perform multiple transactions daily, including balance inquiries, fund transfers, bill payments, and loan applications. The app is expected to handle a large number of users simultaneously, especially during peak hours such as payday or bill due dates. During these periods, a high volume of users may perform different types of transactions, such as large fund transfers or bill payments. Additionally, users often switch between various functions like checking balances, paying bills, and applying for loans. 15
(CO3)
(PO2)
- Given this scenario, which performance testing techniques would you apply to assess the performance of this mobile banking app? Provide detailed reasoning behind your choices and how each technique would help ensure the app performs effectively under different conditions.
6. a) With proper justification, mention the type of maintenance required for the below scenarios: 5 + 5
(CO3)
(PO2)
- i. The cloud-based video streaming service of your company has become increasingly popular among users worldwide. Over time, the platform's user base has expanded, leading to a rise in the volume of video content, user-generated uploads, and simultaneous streams. As a result, the development team notices a gradual decline in streaming quality and buffering issues during peak usage times. Upon investigation, they discovered that Content Delivery Network (CDN) and video encoding methods, which were once adequate, are now unable to handle the high demand for real-time video delivery efficiently.
- ii. The cloud-based Customer Relationship Management (CRM) platform of your company is used by businesses of various sizes. Over the years, the platform has seen steady growth in its user base and data volume. While the platform has been functioning well, the development team has identified that certain underlying components, such as security protocols, data backup systems, and software dependencies, are becoming outdated. Although these components are still functioning, there is a growing risk that they may fail to meet future demands, such as emerging security threats or increased data storage requirements.
- b) During a software upgrade, the testing team is responsible for performing both sanity and regression testing. Discuss the difference between sanity and regression testing, and explain how these testing types contribute to ensuring the stability of the software after new changes. 10
(CO3)
(PO2)
- c) How does continuous delivery differ from continuous deployment? 5
(CO1)
(PO1)

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CSE 4551: Computer Graphics and Multimedia Systems

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1. An infinite double cone in $\mathbb{R}^3$ space, as depicted in Figure 1, with its apex (center) at the point C , axis direction V (used for orientation), and the opening half-angle θ can be represented with the implicit vector Equation 1.

$$H(P) = [(P - C) \cdot V]^2 - (P - C) \cdot (P - C) \cos^2 \theta = 0 \quad (1)$$

A ray in $\mathbb{R}^3$ space can be represented using the parametric explicit equation, $P(t) = R_0 + t \times R_d$, where R_0 is a point, denoting the origin of the ray, and R_d is a vector, denoting the direction of the ray.

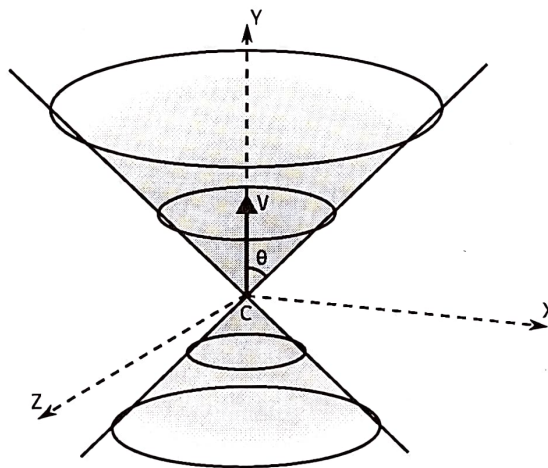


Figure 1: Diagram of an infinite double cone for Question 1

- | | |
|---|----------------------|
| a) Formulate and derive a mathematical model to facilitate calculation of ray-cone intersection from the aforementioned scenario to integrate them in a ray tracer using vector mathematics. You need to formulate vector equations for finding out the value of t at the intersection point and the coordinates of intersection, P . | 10
(CO2)
(PO2) |
| b) To create a single-cone from a double-cone, points need to be restricted to only those that satisfy $(P - C) \cdot V \geq 0$. Similarly, to create a cone of finite height, h , points must be restricted to only those that satisfy $(P - C) \cdot V < h$. Now, based on your formulation in Question 1.a and the aforementioned information, design the pseudo-code of a finite single-cone for a ray tracer. You do not need to calculate the hit normal in the hit function. | 14
(CO4)
(PO3) |
| c) If your ray tracer is not calculating the hit-normal for a cone, what kind of limitations will you face in the rest of the processing pipeline? Explain with an appropriate example. | 6
(CO2)
(PO2) |

2. a) Suppose you are creating a game where an enemy shark AI will be patrolling a body of water and the player needs to cross it safely. There will be fixed patrolling routes based on the position in the body of water and any shark placed will follow the predetermined routes. Based on the aforementioned information, which kind of particle system can be used to simulate the patrolling of a shark? Give appropriate reasoning behind your answer and represent your plan with a diagram. 8
(CO3)
(PO2)
- b) “Combination of curves, randomness, and particle systems can be used to create interesting visual effects” — prove this statement by proposing the design of a visual effect combining the aforementioned concepts. 10
(CO4)
(PO3)
- c) Suppose a viewport is defined with the bottom-left point as (0, 0) and the top-right point as (500, 500). Three lines as defined in Table 1 are being displayed in the viewport. 8
(CO1)
(PO1)

Table 1: Line definitions for Question 2.c

Line	Starting Point	Ending Point
A	(200, 200)	(600, 250)
B	(200, 550)	(550, 650)
C	(300, 300)	(150, 200)

Step-by-step explain with a diagram how each of the lines will be processed for clipping recursively by the Cohen-Sutherland algorithm following the *ABRL* order. You do not need to explicitly calculate any clipped coordinate.

- d) Between Cohen-Sutherland and Cyrus-Beck line clipping algorithms, which one is more suitable for a modern hardware? Explain with appropriate reasoning. 4
(CO1)
(PO1)
3. a) Consider that a CPU based render farm is rendering a huge scenery with vast amount of objects using a custom ray-tracer. However, all the objects are static in the scene and the final output is a single image. There are both sparse and densely populated areas in the scene. What kind of acceleration technique among Boundary Volume Hierarchy (BHV), K-D tree, and Ray Marching Uniform Grid approach will be most suitable in this case? Explain through appropriate comparison between the three approaches based on the use case. 10
(CO3)
(PO2)
 - b) How does explosion of rays occur in recursive ray tracing? Explain with an example ray tree diagram. 8
(CO1)
(PO1)
 - c) How does the depth of recursion affect the output and the performance of a recursive ray tracer? Give proper justification behind your answer. 4
(CO1)
(PO1)
 - d) Among the three optimization approaches mentioned in Question 3.a, which one is most commonly used with modern GPUs and why? Explain with appropriate reasoning. 8
(CO1)
(PO1)

4. a) Imagine that you are rendering a scene of a car driving through a road during night time from the perspective of the driver. The only sources of light in the scene are the car headlights. You have initially used Phong shading model for the road, but you are noticing some visual artifacts related to the lighting. 8 + 8
(CO3)
(PO2)
- i. With appropriate mathematical reasoning and a diagram, explain why the artifacts are occurring.
 - ii. Propose and mathematically explain a solution with a diagram to mitigate the artifact observed in Question 4.a.
- b) How do concepts from statistics help us model micro structures of surfaces in material level? Explain from the perspective of microfacet theorem. 7
(CO1)
(PO1)
- c) How can game engines and real-time renderers make a low-poly geometry appear relatively smooth without actually modifying the geometry itself? Give proper reasoning behind your answer. 7
(CO1)
(PO1)
5. a) A triangle in $\mathbb{R}^3$ space is defined with the three vertices a , b , and c , and a ray is defined with the parametric explicit equation, $P(t) = R_0 + t \times R_d$, where R_0 is a point, denoting the origin of the ray, and R_d is a vector, denoting the direction of the ray. Derive the mathematical expressions to find out the intersection parameter t between the ray and the triangle, and the barycentric coordinates of the intersection, α , β , and γ . 10
(CO1)
(PO1)
- b) Why is barycentric coordinate based ray-triangle intersection more useful than normal based ray-triangle intersection in computer graphics? Briefly explain with two example use cases where barycentric coordinates of intersection are helpful. 7
(CO1)
(PO1)
- c) How can object oriented design principles help in the design of a ray tracer? Explain with an example in support of your answer. 7
(CO1)
(PO1)
- d) Based on your everyday personal experience, list down four cases where you have seen the concepts of computer graphics being used extensively. 6
(CO1)
(PO1)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF NATURAL SCIENCES

Semester Final Examination

Winter Semester: 2023-2024

Course Number: Math 4541

Full Marks: 150

Course Title: Multivariable Calculus and
Complex Variables

Time: 3 Hours

Please answer according to the order of the questions. Answer all the 5 (Five) questions. The symbols have their usual meanings. Marks of each question and the corresponding CO and PO are written in the brackets.

1. a) Find the directional derivative of at $(1, 2)$ in the direction of [10]
 $\vec{u} = \left(\cos \frac{\pi}{3} \right) \hat{i} + \left(\sin \frac{\pi}{3} \right) \hat{j}$ (CO2)
(PO1)
- b) Find an equation of the tangent plane to the hyperboloid [10]
 $z^2 - 2x^2 - 2y^2 = 12$ at the point $(1, -1, 4)$. (CO2)
(PO1)
- c) Determine the absolute maximum and minimum values of [10]
 $f(x, y) = 2 + 2x + 2y - x^2 - y^2$ (CO2)
(PO1)
on the triangular region in the first quadrant bounded by the lines
 $x = 0, y = 0, y = 9 - x$
2. a) A rectangular box is resting on the xy -plane with one vertex at the origin. The [10]
opposite vertex lies in the plane $6x + 4y + 3z = 24$ shown in **Figure Q2.a)**. (CO3)
Find the maximum volume of the box. (PO2)

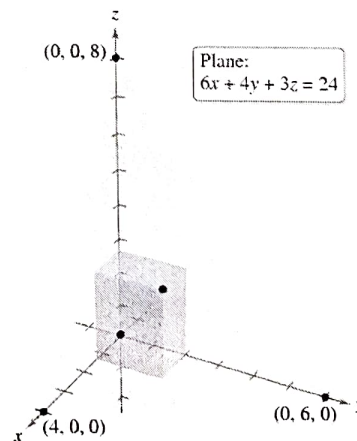


Figure: 2.a)

- b) Determine the minimum value of $f(x, y, z) = 2x^2 + y^2 + 3z^2$ subject to [10]
the constraint $2x - 3y - 4z = 49$. (CO2)
(PO1)

- c) The total weekly revenue that Fostex company realizes in producing and selling its loudspeaker system is given by $R(x, y) = -\frac{1}{4}x^2 - \frac{3}{8}y^2 - \frac{1}{4}xy + 300x + 240y$, where x denotes the number of fully assembled units and y denotes the number of kits produced and sold each week. The total weekly cost attributable to the production of these loudspeakers is $C(x, y) = 180x + 140y + 5000$. Determine how many assembled units and how many kits should be produced per week to maximize the profits. [10] (CO3) (PO2)
3. a) The operators of the Viking Princess, a luxury cruise liner, are contemplating the addition of another swimming pool to the ship. The chief engineer has suggested that an area of the form of an ellipse located in the rear of the promenade deck would be suitable for this purpose. It has been determined that the shape of the ellipse may be described by the equation
$$x^2 + 4y^2 = 3600$$
 where x and y are measured in feet. Determine the dimensions of the rectangular pool with the largest possible area that would meet these requirements. [10] (CO3) (PO2)
- b) Find the average value of $F(x, y, z) = xyz$ throughout the cubical region D bounded by the coordinate planes and the planes $x = 2$, $y = 2$, and $z = 2$ in the first octant. [10] (CO3) (PO2)
- c) Evaluate the volume of region D enclosed by the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$. [10] (CO2) (PO1)
4. a) Find the center of mass of a solid of constant density ρ bounded below by the disk $R: x^2 + y^2 \leq 4$ in the plane $z = 0$ and above by the paraboloid $z = 4 - x^2 - y^2$. [10] (CO3) (PO2)
- b) A thin plate covers the triangular region bounded by the x -axis and the lines $x = 1$ and $y = 2x$ in the first octant. The plate density at the point (x, y) is $\rho(x, y) = 6x + 6y + 6$. Find the plate moments of inertia about the coordinate axes and the origin. [10] (CO3) (PO2)
- c) Evaluate
$$\int_0^3 \int_0^{4(y/2)+1} \int_{y/2}^{\left(\frac{2x-y}{2} + \frac{z}{2}\right)} dx dy dz$$
 by applying the Jacobian transformation and integrating over an appropriate region in uvw -space. [10] (CO2) (PO1)
5. a) Find the work done by the force field $\vec{F} = (y - x^2)\hat{i} + (z - y^2)\hat{j} + (x - z^2)\hat{k}$ along the curve $\vec{r}(t) = t\hat{i} + t^2\hat{j} + t^3\hat{k}$, $0 \leq t \leq 1$, from $(0, 0, 0)$ to $(1, 1, 1)$. [10] (CO3) (PO2)
- b) Using Green's theorem find the outward flux of the vector field $\vec{F}(x, y) = x\hat{i} + y^2\hat{j}$ across the square bounded by the lines $x = \pm 1$ and $y = \pm 1$. [10] (CO3) (PO2)
- c) A fluid of constant density rotates around the z -axis with velocity $\vec{F} = \omega(-y\hat{i} + x\hat{j})$, where ω is a positive constant called the *angular velocity* of the rotation. Find $\nabla \times \vec{F}$ and using Stokes' theorem determine the circulation density. [10] (CO3) (PO2)